

# JIEJIA LIU

(last updated October 9, 2026)

## PERSONAL INFORMATION

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E307, Physics Building  
Tsinghua University  
Beijing, 100084, China

 [jiejialiuhub](https://github.com/jiejialiuhub)  
 [liujj21@mails.tsinghua.edu.cn](mailto:liujj21@mails.tsinghua.edu.cn)  
 0009-0003-8288-3985

Research interests:

Diffuse X-ray emission, Superbubbles, High-resolution spectroscopy, Instrumentation, Galaxy ecosystems.

## EDUCATION

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*Ph.D. in Astronomy & Astrophysics*  
Department of Astronomy, Tsinghua University  
Advisor: Prof. Wei Cui

2021 – expected December 2026

*Bachelor of Physics*  
School of Physics, Nankai University

2017 – 2021

## PUBLICATIONS

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### First Author

1. **Jiejia Liu**, Rui Huang, Chengzhe Li, et al., “X-rays from a newly discovered superbubble in M31”, 2024, [MNRAS](#), **533**, 3027.
2. **Jiejia Liu**, Sifan Wang, Hai Jin, et al., “Preliminary design of detector assembly for DIXE”, 2024, [JLTP](#), **216**, 273, [arXiv:2405.07559](#).
3. **Jiejia Liu**, Chunyang Jiang, Junjie Mao, et al., “Diffuse X-ray Explorer (DIXE): Sky survey strategy and collimator response demodulation”, 2026, [RAA](#), **26**, 125020, [arXiv:2606.10712](#).
4. **Jiejia Liu**, Junjie Mao, Shuinai Zhang, et al., “X-ray measurements of the elemental abundances for the diffuse emission in the nuclear starburst galaxy NGC 3079”, 2026, A&A accepted, [arXiv:2609.37646](#).
5. **Jiejia Liu**, Junjie Mao, Lei Sun, “In-flight validation of the wavelength-offset scaling relation for the XMM-Newton Reflection Grating Spectrometer”, in preparation.

### Co-Author

1. Hai Jin, Junjie Mao, . . . , **Jiejia Liu**, et al., “Diffuse X-ray Explorer: A high-resolution X-ray spectroscopic sky surveyor on the China Space Station”, 2024, [JLTP](#), **215**, 256, [arXiv:2406.09813](#).
2. Ruixuan Tian, Junjie Mao, **Jiejia Liu**, et al., “Simulation of non X-ray background for the DIffuse X-ray Explorer (DIXE) mission”, 2026, [Exp. Astron.](#), **61**, 17, [arXiv:2604.13569](#).
3. Yu Zhou, **Jiejia Liu**, Volodymyr Takhistov, Kazuhisa Mitsuda, “Probing Axion-Photon Mixing with High-resolution X-Ray Spectroscopy”, 2026, [ApJ](#), **1009**, 233, [arXiv:2507.22006](#).
4. Michael C. H. Yeung, Martin G. F. Mayer, . . . , **Jiejia Liu**, et al., “The SRG/eROSITA diffuse soft X-ray background II. spectra and morphology of the eROSITA bubbles in the western Galactic hemisphere”, 2026, A&A accepted, [arXiv:2605.02998](#).
5. Jiayi Chen, Haiwu Pan, . . . , **Jiejia Liu**, et al., “Modeling and in-flight characterization of the point spread function for the Einstein Probe Wide-field X-ray Telescope”, RAA accepted.

## APPROVED PROPOSALS

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Co-I, 300 ks on Einstein Probe Cycle 2 (ID: 0102) 2025  
Title: The Largest Hot Superbubbles Ever Seen in a Low-Density Environment?  
PI: Dr. Rui Huang

## PRESENTATIONS

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### Talks

*X-rays from Superbubble Candidates in M31* 2023.12  
The 2nd HUBS Workshop, Haikou, China  
*Exposure Map and Point-Spread Function of DIXE* 2024.12  
The 2nd ShenKuo Symposium, Xiamen, China  
*X-rays from the superbubble in M31: a candidate for positive stellar feedbacks* 2025.03  
Science of Diffuse Sources in the New Era, Nanjing, China  
*DIXE collimator response and algorithm implementation of DIXE imaging* 2025.06  
Surveying the Hot Baryons Within the Milky Way ISSI-BJ Forum, Beijing, China

### Posters

*X-rays from Superbubble Candidates in M31* 2023.05  
The 26th Guo Shoujing Conference, Huangshan, China  
*X-rays from Superbubble Candidates in M31* 2023.12  
"Resolving Galaxy Ecosystem Across All Scales", Hong Kong, China  
*Preliminary Design of Detector Assembly for DIXE* 2023.06  
The 20th International Conference on Low Temperature Detectors, Daejeon, South Korea  
*Simulation of Magnetic Shielding for DIXE* 2024.06  
SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan

## TEACHING EXPERIENCES

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Teaching Assistant:  
Major course, *Student Seminar* 2023  
Major course, *Radiative Process in Astrophysics* 2024  
Elective course, *Frontiers of Astrophysics* 2025

## SCHOLARSHIPS & AWARDS

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- Comprehensive Scholarship, First Prize, Tsinghua University 2025
- Comprehensive Scholarship, Second Prize, Tsinghua University 2022, 2024
- AMD Scholarship, Tsinghua University 2023
- "Gong Neng" Scholarship, Nankai University 2020
- Academic Excellence Scholarship, Nankai University 2018, 2019

## REFERENCES

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**Prof. Wei Cui** at Tsinghua University

Email: [cui@tsinghua.edu.cn](mailto:cui@tsinghua.edu.cn)

**Prof. Junjie Mao** at Tsinghua University

Email: [jmao@mail.tsinghua.edu.cn](mailto:jmao@mail.tsinghua.edu.cn)

**Prof. Zhijie Qu** at Tsinghua University

Email: [quzhijie@tsinghua.edu.cn](mailto:quzhijie@tsinghua.edu.cn)

**Dr. Yu Zhou** at National Astronomical Observatories

Email: [zhouyu@bao.ac.cn](mailto:zhouyu@bao.ac.cn)